Uncertainty-Based Dendritic Model for Multimodal Remote Sensing Data Classification

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***Abstract*—Multimodal remote sensing data is inevitably affected by noise due to atmospheric conditions, sensor limitations, and other factors. However, existing deep learning-based multimodal remote sensing classification (MRSC) methods overlook the impact of these data noise, which produces the uncertainty and decreases classification accuracy. To address this problem, this article first explores uncertainty-based dendritic model (UDM) for MRSC, which reduces the uncertainty at single-modality feature extraction and multimodal feature fusion stages. At the single-modality feature extraction stage, dendrites, as a novel type of neurons, have been demonstrated with strong feature extraction abilities. Inspired by the dendritic structure, we first design a dendritic-based spatial–channel feature extraction (DSCE) module. Specifically, a dendritic neural layer (DNL) is designed in DSCE. The proposed DNL constructs a multibranch fusion strategy to enhance the expressive capacity of multimodal remote sensing feature extraction, which achieves the localized subspace computation ability. Furthermore, based on the extracted features by DSCE, a dendritic uncertainty-based feature enhancement (DUFE) module is explored to reduce the uncertainty from the extracted features. DUFE exploits uncertainty estimation to adaptively refine the extracted representations, thereby improving feature robustness and discriminative power for MRSC. At the multimodal feature fusion stage, considering the feature redundancy of the different modalities, a dendritic-based uncertainty-aware fusion (DUAF) module is proposed. DUAF performs feature fusion by dynamically assigning weights based on the estimated uncertainty of each modality, thus enhancing classification performance. Experiments on benchmark datasets demonstrate that the proposed UDM outperforms current state-of-the-art (SOTA) methods based on Transformer, convolutional neural network, and Mamba for MRSC. The code is available at https://github.com/hx0558/UDM**

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I. Introduction

REMOTE sensing image classification plays a pivotal role in practical applications, such as smart agriculture [1], disaster monitoring [2], and urban development [3]. Advancements in remote sensing imaging technologies have led to the availability of extensive multimodal datasets, which include several single modalities, such as hyperspectral image (HSIs), light detection and ranging (LiDAR), and synthetic aperture radar (SAR) data. The integration of these single modalities can provide complementary information. For instance, HSI is characterized by their rich spectral information, which enables detailed identification of materials and objects based on their unique spectral signatures [4]. LiDAR offers precise height information and SAR uses microwave pulses to illuminate the target surface and captures the backscattered signal to generate images. This allows SAR to operate regardless of weather conditions or day–night cycles [5].

To maximize the utility of multimodal remote sensing data and improve classification accuracy, two crucial steps must be considered: feature extraction and feature fusion. Predominantly, feature extraction techniques are centered on single modality (i.e., HSI, LiDAR, and SAR), which facilitates the derivation of unique and complementary features that can be subsequently integrated through various fusion approaches [6], [7], [8], [9], [10]. In particular, deep learning-driven methods (i.e., CNN, Transformer, and Mamba) have garnered significant attention in the realm of feature extraction. Specifically, based on the CNN, researchers focused on expanding the width and depth of the architecture [11], [12]. However, this approach places a considerable burden on computational and storage resources. To mitigate this issue, solutions such as dilated convolution and pruning methods within CNN have been proposed for remote sensing classification [13], [14].

In recent years, Transformer models, with their self-attention mechanisms, excel at capturing long-range dependencies within the multimodal remote sensing data. For example, Ding et al. [15] proposed a global–local transformer network to jointly perform LiDAR and HSI classification. Apart from using Transformer models independently, they are often combined with CNNs to improve the classification performance of multimodal data. This integration leverages the strengths

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of both architectures, where CNNs are adept at extracting local spatial features, while Transformers excel at capturing global dependencies and long-range interactions. For instance, Zhang et al. [16] explored a multimodal transformer network to extract both unique and common features of different modalities. Furthermore, Yao et al. [17] proposed a cross-modality attention in combination with CNN for enhancing classification performance.

More recently, Mamba models based on selective state space models have emerged as a promising alternative to Transformer models, offering linear time complexity. For instance, Cao et al. [18] proposed a CLIP-driven Mamba model for multimodal remote sensing classification (MRSC), which delivers high computational efficiency. Similarly, Gao et al. [19] explored a multiscale feature fusion state space model. They designed a spatial Mamba block and a spectral Mamba block to extract unique features from different modalities, attaining linear complexity [19]. Zhang et al. [20] proposed an efficient Mamba-based framework for hyperspectral and LiDAR joint classification, which incorporates a dual-branch siamese visual state space block to extract distinctive features from both modalities.

For MRSC, the process of feature fusion emerges as a pivotal step following feature extraction [21], [22], [23], [24], [25]. Feature fusion methodologies can primarily be categorized into two distinct approaches. The first approach involves initially extracting features from various modalities and subsequently employing straightforward fusion techniques, such as concatenation [26]. The second approach entails modifying the model's architecture and developing specialized fusion strategies. While concatenation strategies offer a relatively simple means of achieving better performance for MRSC, they are inherently characterized by their simplicity in terms of feature fusion. In recent times, researchers have increasingly focused on the latter approach to facilitate more effective information fusion. For instance, the cross-attention mechanism within the Transformer framework has garnered attention. This mechanism computes attention scores between tokens of different modalities, thereby capturing their interdependencies [27]. This enables the model to discern the relationships and dependencies between diverse modalities, leading to the efficient integration of information from different modalities. Recently, the Mamba-based methods fused various modalities by changing the scanning strategies for MRSC [10], [18], [19], [28].

In recent years, inspired by dendritic structures, a new class of networks known as dendritic neural networks has been developed. These models are designed to enhance the representational capacity and overall performance of neural networks, and have been successfully applied to tasks such as time series forecasting and classification [29], [30], [31], [32], [33]. Dendritic principles have also attracted growing attention within the deep learning community. Specifically, Zhang et al. [34] proposed the integration of dendritic learning into vision Transformer, improving the classification performance. Liu et al. [35] introduced dendritic deep supervision neural networks to strengthen the network's capacity for nonlinear feature mapping.

However, the aforementioned MRSC approaches overlook the inherent uncertainty introduced by data noise. In deep learning, uncertainty is generally categorized into aleatoric uncertainty and epistemic uncertainty. Aleatoric uncertainty arises from the inherent characteristics of the data, such as sensor noise, atmospheric conditions, or spatial heterogeneity [36], whereas epistemic uncertainty stems from limited knowledge or insufficient training samples. During the acquisition of different modalities, diverse noises are inevitably introduced owing to sensor-specific characteristics, imaging conditions, and transmission disturbances [37]. Moreover, uncertainties also arise in the fusion process because of the inherent discrepancies among the feature spaces of the individual modalities [38]. Furthermore, ignoring uncertainty during the fusion process can result in redundant feature representations. In the field of MRSC, uncertainty exists in both the feature extraction and feature fusion stages. On the one hand, current feature extraction strategies neglect such uncertainty. On the other hand, in the feature fusion stage, uncertainty often introduces information redundancy. Since epistemic uncertainty can be mitigated by increasing the amount of data, this study focuses on aleatoric uncertainty.

To address the challenges posed by noise and uncertainty in multimodal remote sensing data, this article proposes an uncertainty-based dendritic model (UDM) to enhance the uncertainty awareness across the feature extraction and the feature fusion stages. Unlike standard Bayesian neural networks that primarily target epistemic uncertainty by modeling distributions over model weights [39], our method focuses on modeling aleatoric uncertainty in the feature space. This design choice aims to explicitly mitigate the inherent sensor noise in heterogeneous remote sensing modalities. Due to the complex acquisition conditions of multimodal remote sensing data, existing deep learning-based classification methods often overlook the adverse impact of data noise, leading to increased uncertainty and degraded classification accuracy. To tackle these problems, the proposed UDM systematically incorporates dendritic-inspired computation and uncertainty quantification into all stages of the model pipeline, achieving progressive uncertainty reduction and robust multimodal representation learning. Specifically, to realize uncertainty-aware feature representation and build a reliable foundation for subsequent modules, a dendritic-based spatial–channel feature extraction (DSCE) module is first designed in the single-modality feature extraction. The DSCE module adopts the proposed dendritic neural layer (DNL) that performs branch-specific local computations by projecting inputs onto dendritic branches via independent synaptic weights, modulating them with branch-specific biases, and finally integrating them through membrane-level weighted aggregation. This mechanism enhances spatial–spectral modeling capabilities and yields more discriminative and robust single-modality representations. Based on the extracted features, a dendritic uncertainty-based feature enhancement (DUFE) module is further proposed to refine the learned representations. By estimating and leveraging aleatoric uncertainty, DUFE adaptively emphasizes reliable information while suppressing ambiguous signals. In addition, a contrastive learning strategy is

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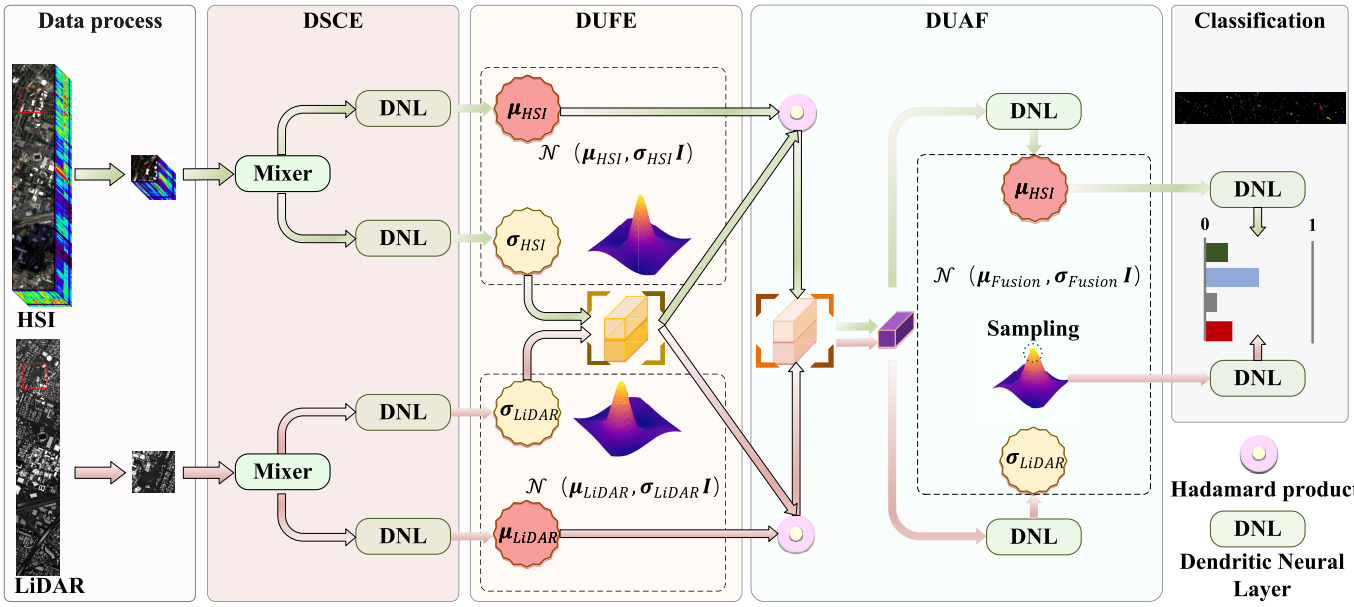


Fig. 1. Illustration of the proposed UDM for multimodal classification. Overall structure is divided into five stages, namely data process, DSCE, DUFE, DUAF, and classification.

introduced to strengthen intramodal feature consistency under uncertainty guidance. At the multimodal feature fusion stage, a dendritic-based uncertainty-aware fusion (DUAF) module is developed. DUAF dynamically integrates multimodal features by assigning adaptive weights according to their estimated uncertainties, thereby achieving interpretable and robust fusion that balances intermodal complementarity and redundancy. The main contributions of this article are summarized as follows.

1) In this article, we first introduce the uncertainty-aware framework dedicated to multimodal remote sensing data classification, where uncertainty perception is systematically realized by jointly incorporating dendritic-inspired mechanisms and uncertainty quantification into the single-modality feature extraction, and multimodal feature fusion stages.
2) At the single-modality feature extraction stage, the DSCE module is first proposed to capture discriminative feature representations. Within DSCE, the proposed DNL serves as the core computational unit, enhancing the spatial–spectral modeling capability and improving the robustness of feature extraction.

Furthermore, building upon the extracted single-modality features, DUFE quantifies these uncertainties and incorporates contrastive learning to refine modality-specific representations, thereby enhancing feature robustness and discriminability under uncertainty.

1) At the multimodal feature fusion stage, considering that direct fusion of features from different modalities may introduce feature redundancy, DUAF module is proposed to specifically address this issue. DUAF dynamically adjusts the fusion weights of different modalities based on their estimated uncertainties, thereby achieving balanced and adaptive multimodal feature fusion.
2) We conducted extensive experiments on three public multimodal remote sensing datasets, i.e., Houston, Berlin, and MUUFL. The experimental results demonstrate the effectiveness of the proposed method. Moreover, ablation experiments further validate the advantages and effectiveness of the feature extraction module, the feature enhancement module, and the feature fusion module. For instance, when employed as the feature extraction module, DNL achieves 11.8% fewer parameters and 9.77% lower FLOPs compared to Transformer.

The overall structure of this article is presented as follows. The background of this study is outlined in Section I. Section II unfolds the proposed method, including the dendritic-based feature extraction module, the feature enhancement module, and the feature fusion module. Experimental settings, classification results, ablation experiments, and parameter analysis are detailed in Section III. Finally, Section IV summarizes this article and provides an outlook on future directions.

II. Methods

A. Overview of the Proposed UDM

In this section, we present the complete architecture of the proposed UDM in Fig. 1. As illustrated in the figure, the framework begins with data preprocessing and is mainly divided into two parts: single-modality feature extraction and multimodal feature fusion. The single-modality feature extraction stage comprises the DSCE and DUFE modules, which perform preliminary feature extraction and feature enhancement, respectively. In the multimodal feature fusion stage, the DUAF module is proposed to dynamically integrate the enhanced features generated by DUFE. Finally, two DNLs are employed as classifiers to achieve the classification task. UDM

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integrates multisource RS data (e.g., HSI, LiDAR, and SAR) by harnessing the feature extraction capabilities of dendritic neurons and an uncertainty-weighted fusion approach.

The task we investigate is pixel-level classification using spatial and spectral information. This article employs HSI and LiDAR/SAR data. For given multimodal data, we define input HSI data as $\boldsymbol{X}_h \in \mathbb{R}^{h\times w\times c}$ and LiDAR/SAR data as $\boldsymbol{X}_l \in \mathbb{R}^{h\times w\times b}$, where h and w denote the height and width of the image, respectively, and c denotes the spectral channel numbers. For each pixel, there is an HSI cube $\boldsymbol{x}_h \in \mathbb{R}^{p\times p\times c}$ obtained from the HSI data $\boldsymbol{X}_h$ and a LiDAR/SAR cube $\boldsymbol{x}_l \in \mathbb{R}^{p\times p\times b}$ obtained from the LiDAR/SAR data $\boldsymbol{X}_l$, where p denotes the patch size, the cube pair as input data for multimodal classification task. The objective is to assign a class label $y \in \{1, 2, \ldots, k\}$ to each pixel, where k denotes the total number of classes.

B. Dendritic-Based Spatial–Channel Feature Extraction

Neural networks have achieved impressive performance across various real-world tasks. However, their reliance on traditional linear layers for dimensional transformations constrains their ability to capture complex and nonlinear feature relationships. Furthermore, existing artificial neurons suffer from inherent limitations, such as weak modeling capability and the lack of structural diversity. In contrast, dendritic neurons, inspired by neuroscience, implement subspace-local computations on multiple dendritic branches and integrate them through multibranch fusion, which enables more flexible and fine-grained feature representation.

Leveraging the inherent biological interpretability of dendritic neurons, this article introduces a promising solution for MRSC. To further enhance the subspace-local computation and multibranch integration capabilities of neural networks, we propose a DNL module, inspired by the operational mechanisms of biological dendritic neurons. This biophysically derived architecture simulates the four-stage information processing hierarchy observed in dendritic systems. Through arborized branch structures implementing localized aggregation, the proposed DNL achieves enhanced representation capability for inputs. Fig. 2 schematically details this architecture. In the context of multimodal remote sensing data, each modality exhibits distinct statistical properties and structural characteristics. Even within a single modality, the data often contain highly complex and heterogeneous information. The proposed DNL addresses this limitation through its multibranch structure. Each branch performs localized transformations, enabling the network to disentangle different types of intramodal features and to capture fine-grained details that would otherwise be overlooked by conventional linear layer.

The proposed DNL treats each neuron as a structured unit consisting of multiple dendritic branches. Each branch receives synaptic inputs from the input vector via a dedicated weight set and performs local integration before being aggregated at the membrane layer. This module is designed based on the structure and function of real neurons, incorporating four key computational stages: the synaptic layer, dendritic layer, membrane layer, and soma layer. Each stage corresponds to a biophysical process in neuronal computation and collectively contributes to robust and flexible feature transformation.

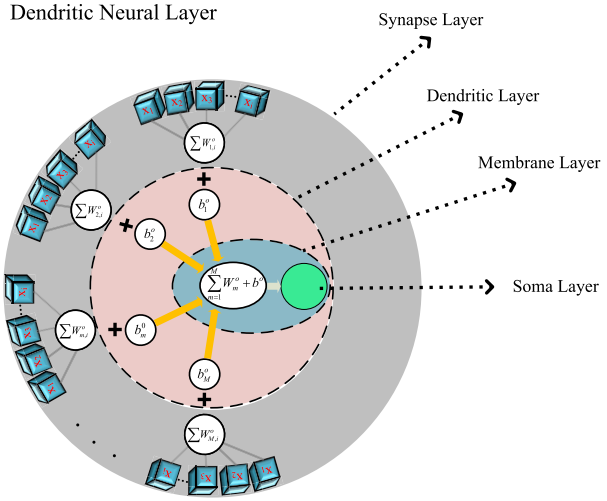


Fig. 2. Structure of the proposed DNL. The dendritic structure is composed of four layers: synaptic layer, dendritic layer, membrance layer, and soma layer. M denotes the number of dendritic branches.

Specifically, given an input vector $\boldsymbol{X} \in \mathbb{R}^I$ with I channels, and assuming each output neuron $o \in \{1, \ldots, O\}$ possesses M dendritic branches. In the synaptic layer, input features are projected onto multiple dendritic branches using independent synaptic weights. The synaptic response of the mth branch of the oth neuron is computed as $\boldsymbol{s}_m^o = \sum_{i=1}^{I} \boldsymbol{W}_{m,i}^{(1)o} \cdot \boldsymbol{x}_i$, where $\boldsymbol{W}_{m,i}^{(1)o} \in \mathbb{R}$ denotes the synaptic weight connecting input channel i to dendritic branch m of neuron o. In the dendritic layer, a branch-specific bias is added to modulate the local excitability of each dendrite, mimicking branch-level thresholds observed in biological neurons. Accordingly, the output of the dendritic layer is given by $\boldsymbol{d}_m^o = \boldsymbol{s}_m^o + \boldsymbol{b}_m^{(1)o}$, where $\boldsymbol{b}_m^{(1)o} \in \mathbb{R}$ is the branch-specific dendritic bias that modulates the local excitability and introduces spatial diversity in feature encoding. In the membrane and soma layers, the signals from all dendritic branches of a neuron are aggregated via a weighted sum to form the membrane potential, which is then finally transmitted to the soma. Specifically, the computation is expressed as $y^o = \sum_{m=1}^{M} \boldsymbol{W}_m^{(2)o} \cdot \boldsymbol{d}_m^o + \boldsymbol{b}^{(2)o}$, where $\boldsymbol{W}_m^{(2)o} \in \mathbb{R}$ represents the membrane-level contribution weight of the mth branch, and $\boldsymbol{b}^{(2)o}$ is a neuron-specific membrane bias, and y^o denotes the final output transmitted to the soma.

The proposed DNL module serves as an alternative to conventional linear layer, aiming to provide structured transformation capabilities. Although the DNL draws inspiration from biological mechanisms, it is designed to be fully compatible with modern deep learning frameworks. Specifically, all operations within the synaptic, dendritic, and membrane layers involve weighted summations and bias additions, which are continuously differentiable functions. Consequently, the standard backpropagation training pipeline remains unchanged. The error gradients flow from the soma layer back to the individual dendritic branches and finally to the synaptic weights

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via the standard chain rule. This ensures that the DNL module can be optimized end-to-end using conventional gradient descent algorithms without any architectural modifications to the solver. By incorporating localized subspace computation and multibranch fusion, DNL enhances the representational capacity beyond basic feature mapping. While nonlinear activation in the soma is not employed in this study, the two-stage design allows the module to realize more complex transformations than a single-layer mapping. The overall output of the DNL for an input $\boldsymbol{X}$ is the vector $\boldsymbol{Y} = [y^1, y^2, \ldots, y^O]$, where each y^o is the soma output of the oth neuron.

To capture spatial–channel features within each modality, we incorporate the proposed DNL into the DSCE module for every modality. DNL structures a single neuron into multiple dendritic branches, allowing different branches to specialize in processing distinct spatial–channel information within local regions and generate high-level local representations. Subsequently, these branch responses are dynamically aggregated at the membrane layer to obtain robust and semantically rich local channel representations. With DNL integrated into DSCE, this module not only enables both nonlinearity and local computation but also delivers strong interpretability through branch visualization, thereby providing more discriminative representations for subsequent feature enhancement and feature fusion.

C. Uncertainty-Based Feature Enhancement

Remote sensing data acquired from different modalities inherently contain various sources of uncertainty due to sensor noise, environmental conditions, and acquisition limitations. To mitigate the impact of these uncertainties, we build upon the extracted features in the proposed DSCE module and propose an uncertainty-based feature enhancement module in this section.

MRSC relies on integrating heterogeneous modalities, including HSI with hundreds of contiguous spectral bands that capture fine-grained material reflectance [40], LiDAR providing spatial elevation information [41], and SAR providing polarimetric information [42]. However, the acquisition principles and physical characteristics of these three types of data inevitably entail information disturbances with uncertainty during their generation. For HSI, during spectral acquisition, the inherent thermal noise of the sensor causes random fluctuations in data of some bands [43]. For LiDAR, multipath effects from multiple surface reflections during laser transmission and reception cause point cloud deviations and reduce spatial accuracy [44]. For SAR, the coherent scattering of electromagnetic waves generates speckle noise, which obscures the texture details of land-cover objects [45]. These information disturbances are not caused by model errors but are inherent to the physical processes of data acquisition and modal characteristics. They directly reduce the feature discrimination capability of models. To mitigate the impact of data uncertainty, we propose a DUFE module. The module comprises Sections II-C1 and II-C2.

1) Uncertainty Quantification in Deep Learning Models for Multimodal: According to [39], uncertainties in MRSC can be categorized into two distinct types with clear boundaries: aleatoric uncertainty and epistemic uncertainty. Aleatoric uncertainty is intrinsic to the data, representing inherent stochasticity that cannot be eliminated regardless of additional information acquisition. All the aforementioned uncertain information disturbances occurring during the data generation process are classified as aleatoric uncertainty. In contrast, epistemic uncertainty originates from the model's incomplete learning of multimodal remote sensing data, tied directly to uncertainty in model parameters, and it can be reduced by increasing training data.

For the given data, we learn distributional representations of the feature rather than a definite point representation in the feature space. To quantify these uncertainties, we perform N iterative forward propagation passes during inference, where dropout-enabled stochastic sampling approximates the true predictive distribution for uncertainty measurement. Here, dropout-enabled stochastic sampling refers to randomly masking part of the model weights in each forward pass, and w_i denotes the ith set of sampled masked weights obtained from this process. By adopting a Gaussian prior distribution $p(y \mid x) \sim \mathcal{N}\left(\mu_*(x), \sigma_*^2(x)\right)$, and further assuming that the model outputs follow a Gaussian distribution $p(y \mid x, w_i) \sim \mathcal{N}\left(\mu_i(x), \sigma_i^2(x)\right)$. We can derive, based on the law of total variance, that

$$\mu_*(x) = \frac{1}{N}\sum_i \mu_i(x) \tag{1}$$

$$\begin{aligned}\sigma_*^2(x) &= \frac{1}{N}\sum_i \sigma_i^2(x) + \frac{1}{N}\sum_i \mu_i^2(x) - \mu_*^2(x) \\ &= \mathbb{E}_i\left[\sigma_i^2(x)\right] + \mathrm{Var}_i\left[\mu_i(x)\right]\end{aligned} \tag{2}$$

where $\mathrm{Var}_i[\mu_i(x)]$ represents epistemic uncertainty, reflecting variability in model outputs caused by fluctuations in model's parameters. While $\mathbb{E}_i[\sigma_i^2(x)]$ quantifies aleatoric uncertainty, capturing the variability caused by the inherent noise in the input data. Notably, our focus in feature enhancement is exclusively on aleatoric uncertainty. This choice is justified by the uncertainty properties defined in [39]. Epistemic uncertainty can be mitigated through increased data acquisition. In contrast, aleatoric uncertainty is inherent to the data, quantifying that it enables the achievement of feature enhancement for the data. Thus, we enhance single-modality features by explicitly quantifying modality-specific aleatoric uncertainties, as detailed in Section II-C2.

2) Single-Modality Feature Enhancement: Based on the aforementioned quantification of aleatoric uncertainty, we propose an aleatoric uncertainty-based method to enhance single-modality feature. As highlighted earlier, our emphasis leans toward aleatoric uncertainty, given its inherent origin in dataset noise. Consequently, we first quantify each aleatoric uncertainty of single-modality before fusion to empower subsequent learning of more discriminative joint features. Given the multimodal samples $\boldsymbol{x}_i^m$, where m denotes different modality, the uncertainty of each modality is quantified by learning its distributional representations. According to (2), aleatoric uncertainty can be predicted via the output variance $\boldsymbol{\sigma}_i^{m2}$ of deep learning models. We first utilize a feature extractor to

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learn the modality-independent embedding, then deploy two dendritic layers dedicated to learning the modality-specific mean vector $\boldsymbol{\mu}_i^m$ and variance vector $\boldsymbol{\sigma}_i^{m2}$. Furthermore, the latent variable $\boldsymbol{z}_i^m$ in the feature space is modeled as a multivariate Gaussian distribution

$$p\left(\boldsymbol{z}_i^m \mid \boldsymbol{x}_i^m\right) \sim \mathcal{N}\left(f_{\theta 1}^m(\boldsymbol{x}_i^m),\ f_{\theta 2}^m(\boldsymbol{x}_i^m)\mathbf{I}\right) \tag{3}$$

where $f_{\theta 1}^m(\cdot)$ and $f_{\theta 2}^m(\cdot)$ denote processing through two dendritic layers, computing the mean vector $\boldsymbol{\mu}_i^m$ and variance vector $\boldsymbol{\sigma}_i^{m2}$, respectively. The mean vector μ_i^m represents a stable feature of each modality sample in the latent space and can be regarded as a denoised core feature. The variance vector $\boldsymbol{\sigma}_i^{m2}$ captures the aleatoric uncertainty representation of each modality. Following the principle of variational autoencoders (VAEs) [46], a standard Gaussian prior $\mathcal{N}(0, \mathbf{I})$ is imposed on the latent variable z_i^m, where $\mathbf{I}$ denotes the identity matrix. To minimize the Kullback–Leibler (KL) divergence between each modality's distribution and the prior Gaussian distribution

$$L_{\mathrm{KLgauss}} = \sum_{m}^{m\in\mathbb{M}} \mathrm{KL}\left(p\left(\boldsymbol{z}_i^m \mid \boldsymbol{x}_i^m\right) \| \mathcal{N}\left(0, \mathbf{I}\right)\right). \tag{4}$$

$$\mathrm{KL}\left(p\left(\boldsymbol{z}_i \mid \boldsymbol{x}_i\right) \| \mathcal{N}\left(0, \mathbf{I}\right)\right) = -\frac{1}{2}\sum_i \left(1 + \log\sigma\left[i\right]^2 - \mu\left[i\right]^2 - \sigma\left[i\right]^2\right). \tag{5}$$

The loss function enables the feature distribution learned by each modality to approach a prior Gaussian distribution.

Meanwhile, the inherent aleatoric uncertainty of data cannot be fully eliminated. By actively simulating such noise to minimize its impact on the final feature representations, we can achieve feature enhancement. For a given unimodal multivariate Gaussian distribution $p(\boldsymbol{z}_i^m \mid \boldsymbol{x}_i^m) \sim \mathcal{N}\left(\boldsymbol{\mu}_i^m, \boldsymbol{\sigma}_i^{m2}\mathbf{I}\right)$, we sample a prototype anchor point $\boldsymbol{z}_i^{m,(a)}$ from this distribution, while stochastically drawing another sample $\boldsymbol{z}_i^{m,(+)}$ from the same distribution as a positive instance. Additionally, we sample a negative sample $\boldsymbol{z}_j^{m,(-)}$ from the distribution of a distinct sample $(i \neq j)$, defined as $p(\boldsymbol{z}_j^m \mid \boldsymbol{z}_j^m) \sim \mathcal{N}\left(\boldsymbol{\mu}_j^m, \boldsymbol{\sigma}_j^{m2}\mathbf{I}\right)$. The anchor and positive pair $(\boldsymbol{z}_i^{m,(a)}, \boldsymbol{z}_i^{m,(+)})$ sampled identically from the same distribution inherently represent noise-perturbed variants derived from identical semantic content under stochastic uncertainty of the modality. In contrast, the negative sample $\boldsymbol{z}_j^{m,(-)}$ originates from entirely different semantics. Building on this framework, we aim to maximize similarity between the anchor and its corresponding positive instance while minimizing similarity between anchor and unrelated negative instances, thereby constructing a contrastive learning for unimodal feature enhancement

$$L_{\mathrm{FE}} = -\log \frac{e^{\left(\frac{\phi\left(\boldsymbol{z}_i^{m,(a)}, \boldsymbol{z}_i^{m,(+)}\right)}{\tau}\right)}}{\sum_{j\neq i}\left(e^{\left(\frac{\phi\left(\boldsymbol{z}_i^{m,(+)}, \boldsymbol{z}_j^{m,(a)}\right)}{\tau}\right)} + e^{\left(\frac{\phi\left(\boldsymbol{z}_j^{m,(-)}, \boldsymbol{z}_i^{m,(a)}\right)}{\tau}\right)}\right)} \tag{6}$$

where $\phi(\boldsymbol{u}, \boldsymbol{v}) = \boldsymbol{u}^T\boldsymbol{v}/\|\boldsymbol{u}\|\|\boldsymbol{v}\|$ denotes cosine similarity and $\tau > 0$ is a temperature coefficient. The loss fusion L_{FE} is designed to pull positive pairs together, encouraging the model to learn that although samples with the same semantics may yield slightly different features under the same modality due to aleatoric uncertainty, they should still be located close to each other in the feature space. This enhances the model's tolerance to inherent data noise. Simultaneously, it repels negative pairs apart to ensure that samples with distinct semantics are decisively separated, preserving feature discriminability. Injecting uncertainty-based noise into the feature space for contrastive learning acts as a strong regularizer, guiding the network to learn smoother and broader decision boundaries while avoiding overfitting to specific noise patterns.

To ensure that the sampling process is differentiable and the model can be trained end-to-end, we apply the reparameterization trick [47], which enables gradients to propagate through both $\boldsymbol{\mu}_i^m$ and $\boldsymbol{\sigma}_i^m$

$$\boldsymbol{z}_i^m = \boldsymbol{\mu}_i^m + \eta\boldsymbol{\sigma}_i^m, \quad \eta \sim \mathcal{N}\left(0, \mathbf{I}\right). \tag{7}$$

D. Uncertainty-Aware Feature Fusion

Although single-modality feature enhancement achieves intramodal stable feature representations by explicitly modeling and leveraging aleatoric uncertainty, the effectiveness of multimodal classification ultimately depends on the fusion quality of these heterogeneous features. Therefore, it is essential to fully exploit complementary information across modalities while simultaneously mitigating the negative impact of uncertainty during the fusion process. Furthermore, since multimodal remote sensing data originate from the same scene, fusion stage may introduce redundant features. To address the aforementioned problems, a DUAF module is proposed, which mainly consists of two components: modality alignment and feature fusion.

First, to preserve semantic consistency across HSI data and SAR/LiDAR data, a loss function based on the KL divergence is designed as follows:

$$L_{\mathrm{KLcross}} = \sum_{m_1\neq m_2}^{m_1,m_2\in\mathbb{M}} \mathrm{KL}\left(p\left(\boldsymbol{z}_i^{m_1} \mid \boldsymbol{x}_i^{m_1}\right) \| p\left(\boldsymbol{z}_i^{m_2} \mid \boldsymbol{x}_i^{m_2}\right)\right) \tag{8}$$

$$\begin{aligned}&\mathrm{KL}\left(p_1\left(\boldsymbol{z}_i \mid \boldsymbol{x}_i\right) \| p_2\left(\boldsymbol{z}_i \mid \boldsymbol{x}_i\right)\right)\\ &= \frac{1}{2}\sum_i \left(\log\frac{\sigma_2^2\left[i\right]}{\sigma_1^2\left[i\right]} + \frac{\sigma_1^2\left[i\right] + \left(\mu_1^2\left[i\right] - \mu_2^2\left[i\right]\right)^2}{\sigma_2^2\left[i\right]} - 1\right).\end{aligned} \tag{9}$$

The loss function concurrently forces the distribution representations of both modalities closer to each other, achieving the semantic consistency.

Although semantic consistency across modalities is ensured, the fusion process remains influenced by uncertainty and tends to produce redundant feature representations. Specifically, different modalities contribute unequally to joint representations [48], making it necessary to adopt a dynamic fusion strategy. Single-modality feature enhancement has already provided stable unimodal representations $\boldsymbol{\mu}_i^m$ and quantified modality uncertainty $\boldsymbol{\sigma}_i^m$. Based on the definition of uncertainty, a smaller $\boldsymbol{\sigma}_i^m$ implies greater stability in the modality's feature

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representation and should thus carry higher weights in the fusion

$$\bar{\boldsymbol{v}}_i^f = \sum_{m\in\mathbb{M}} \alpha_i^m \boldsymbol{\mu}_i^m, \quad \alpha_i^m = \text{softmax}\left(\frac{1}{\boldsymbol{\sigma}_i^m}\right) \tag{10}$$

where $\bar{\boldsymbol{v}}_i^f$ represents the joint distribution representation across m modalities. The dynamic fusion mechanism computes attention weights α_i^m based on the aleatoric uncertainty $\boldsymbol{\sigma}_i^m$ of each modality, enabling the mitigation of the adverse effects of uncertainty during feature integration.

To further address information redundancy in fused features, we employ the variational information bottleneck (VIB) theory [49] to compress the joint distribution representation $\bar{\boldsymbol{v}}_i^f$, utilizing two dendritic layers to generate mean vector and variance vector projected into a latent Gaussian distribution

$$\begin{aligned} p(\bar{\boldsymbol{z}}_i \mid \bar{\boldsymbol{v}}_i^f) &\sim \mathcal{N}\left(\bar{\boldsymbol{\mu}}_i, \bar{\boldsymbol{\sigma}}_i^2 \mathbf{I}\right) \\ \bar{\boldsymbol{\mu}}_i = f_{\theta 1}(\bar{\boldsymbol{v}}_i^f), &\quad \bar{\boldsymbol{\sigma}}_i = f_{\theta 2}(\bar{\boldsymbol{v}}_i^f) \end{aligned} \tag{11}$$

where $f_{\theta 1}(\cdot)$ and $f_{\theta 2}(\cdot)$ denote two dendritic layers. The reparameterization trick is likewise employed here. The objective function is minimized as follows:

$$\begin{aligned} L_{\text{Fusion}} = \lambda \text{KL}\left(p\left(\bar{\boldsymbol{z}}_i \mid \bar{\boldsymbol{v}}_i^f\right) \| \mathcal{N}(0, \mathbf{I})\right) \\ -y^T \log\left(\text{softmax}\left(f_\theta\left(\bar{\boldsymbol{z}}_i\right)\right)\right) \end{aligned} \tag{12}$$

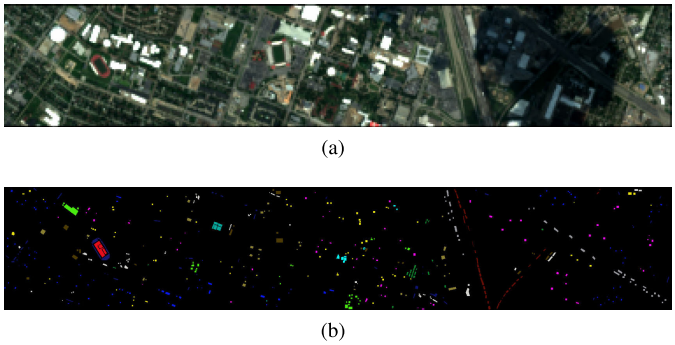
where λ is a hyperparameter, and $f_\theta(\cdot)$ denotes the dendritic layer used for classification. While preserving classification accuracy, the KL divergence regularization term constrains the latent space distribution to approximate a standard Gaussian prior, thereby effectively suppressing the introduced cross-modal redundant information.

III. Experiments

In this section, we present the experimental datasets and configurations employed in this article. In addition, ablation studies and comparative experiments were conducted to demonstrate the effectiveness and superiority of the proposed method.

A. Datasets Description

1) Houston Dataset: The Houston Hyperspectral-LiDAR dataset, jointly developed by the University of Houston and the NSF-funded Center for Airborne Laser Mapping, integrates co-registered HSI (144 spectral bands spanning 380–1050 nm at 2.5 m spatial resolution, acquired via the ITRES CASI-1500 sensor) with LiDAR-derived elevation data (2.1 points/m^2 density, 0.5 m vertical RMSE accuracy). The dataset exhibits spatial dimensions of 349 × 1905 pixels, containing 15 029 ground truth samples across 15 distinct classes. For experimental validation, 200 samples were randomly selected as the training subset, with the remaining 14 829 samples allocated for testing. The dataset content is illustrated in Fig. 3, while the detailed statistics including class names and sample sizes are summarized in Table I.



(a)

(b)

Fig. 3. (a) False color composite of the Houston dataset (bands 63-40-23). (b) Ground reference map with 15 land-cover classes.

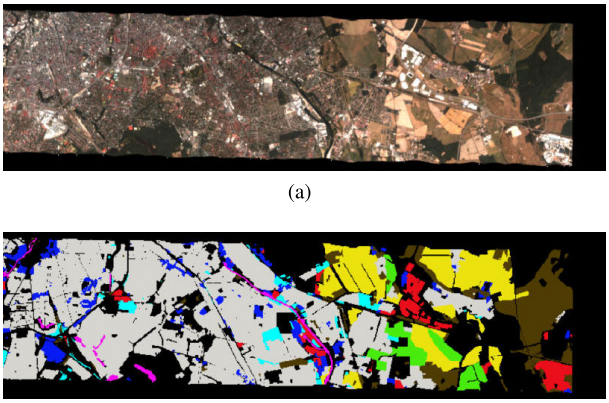
TABLE I
Label Information of the Houston Dataset

No.	Color	Name	Numbers
1		Healthy Grass	1251
2		Stressed Grass	1254
3		Synthetic Grass	697
4		Trees	1244
5		Soil	1242
6		Water	325
7		Residential	1268
8		Commercial	1244
9		Road	1252
10		Highway	1227
11		Railway	1235
12		Parking Lot 1	1233
13		Parking Lot 2	469
14		Tennis Court	428
15		Running Track	660
Total			**15029**

2) Berlin Dataset: The Berlin multimodal remote sensing dataset contains co-registered HSI and SAR data covering Berlin, Germany. The hyperspectral data comprises 244 spectral bands spanning 400–2500 nm with 797 × 220 pixel dimensions, capturing detailed geometric-spatial information and fine-grained spectral characteristics of urban surfaces. The complementary SAR data encompass four channels with identical pixel dimensions as the HSI. This dataset contains 464 671 annotated pixels distributed across 8 land-cover classes. An overview of the dataset is illustrated in Fig. 4. Hyperspectral-SAR pixel-level registration ensures precise multimodal feature alignment for joint spectral–polarimetric urban monitoring applications. Table II summarizes the specific land-cover class information.

3) MUUFL Dataset: The MUUFL dataset, acquired in November 2010 on the University of Southern Mississippi Gulfpark Campus (Long Beach, MI, USA), contains synchronously acquired HSI and LiDAR data with spatial co-registration. The hyperspectral data

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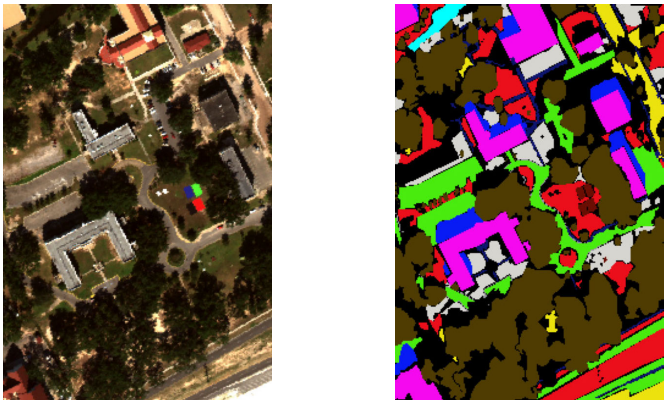


(a) (b)

Fig. 4. (a) False color composite of the Berlin dataset (bands 37-19-3). (b) Ground reference map with 8 land-cover classes.

TABLE II
LABEL INFORMATION OF THE BERLIN DATASET

No.	Color	Name	Numbers
1		Forest	51954
2		Residential Area	268642
3		Industrial Area	19566
4		Low Plants	59282
5		Soil	17426
6		Allotment	13305
7		Commercial Area	24824
8		Water	6672
Total			**464671**



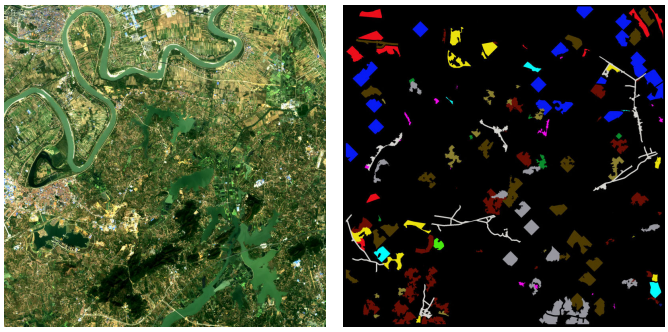
(a) (b)

Fig. 5. (a) False color composite of the MUUFL dataset (bands 25-14-6). (b) Ground reference map with 8 land-cover classes.

originally comprised 72 spectral bands (375–1050 nm) captured at 325 × 220 pixels as shown in Fig. 5, though standard protocols remove the first 4 and last 4 noise-affected bands, retaining 64 validated spectral channels. The dataset provides 53 687 ground truth samples distributed across 11 land-cover classes, including urban and vegetated terrain categories. The detailed information is reported in Table III.

TABLE III
LABEL INFORMATION OF THE MUUFL DATASET

No.	Color	Name	Numbers
1		Trees	23246
2		Mostly Grass	4270
3		Mixed ground surface	6882
4		Dirt and sand	1826
5		Road	6687
6		Water	466
7		Buliding shadow	2233
8		Building	6240
9		Sidewalk	1385
10		Yellow curb	183
11		Cloth panels	269
Total			**53687**



(a) (b)

Fig. 6. (a) False color composite of the Wuhan dataset. (b) Ground reference map with 13 land-cover classes.

4) Wuhan Dataset: The Wuhan multimodal remote sensing dataset, derived from the C2Seg-BW collection, integrates coregistered data acquired by the Gaofen-5 and Gaofen-6 satellites over Wuhan, China. The dataset comprises 116 validated hyperspectral bands spanning the 400–2500 nm range and 4 complementary multispectral bands, both aligned with a standardized spatial resolution of 10 m. For experimental validation, a representative region was selected, which contains 2000 × 2000 pixels, providing 509 996 labeled ground truth samples distributed across various land-cover classes. The dataset content is illustrated in Fig. 6, while the detailed statistics including class names and sample sizes are summarized in Table IV.

B. Experimental Setting

1) Configuration: The experiments were conducted on an NVIDIA GeForce RTX 4060 Ti GPU using PyTorch with the Adam optimizer. During data preparation, input HSI data are processed into $27 \times 27 \times C$ pixel patches and other sources of multimodal data are processed into $27 \times 27 \times B$ pixel patches. Empirically determined parameters included a batch size of 100 and 300 training epochs, with initial learning rates set to 1×10^{-4} for the Houston and MUUFL datasets, and

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TABLE IV
LABEL INFORMATION OF THE WUHAN DATASET

No.	Color	Name	Numbers
1		Surface water	98462
2		Street network	47943
3		Urban fabric	32866
4		Industrial, commercial, and transport	35423
5		Mine, dump, and construction sites	3088
6		Artificial vegetated areas	13246
7		Arable land	93675
8		Permanent crops	5775
9		Pastures	251
10		Forests	65606
11		Shrub	87678
12		Open spaces with no vegetation	20737
13		Inland wetlands	5245
Total			**509995**

5×10^{-5} for the Berlin and Wuhan datasets. To ensure fairness, 200 samples were randomly selected from each dataset as the training set, while the remaining samples were used as the test set. To ensure stable convergence, a step learning rate scheduler was employed (decay rate set to 0.1) to fine-tune the optimization. To facilitate reproducibility, we further specify the internal structural parameters. The feature embedding dimension was configured to 256 for the Houston, MUUFL, and Wuhan datasets, whereas a compact dimension of 128 was adopted for the Berlin dataset. The proposed DNL was instantiated with 2 dendritic branches per neuron to enable diverse subspace computations. For uncertainty quantification, we performed 10 stochastic sampling iterations during forward propagation to approximate the predictive distribution.

2) Evaluation Metrics: The model performance is evaluated through three metrics: average accuracy (AA), Kappa coefficient, and overall accuracy (OA). OA represents the proportion of correctly classified instances across the entire test set, whereas AA computes the mean of per-class accuracies to mitigate class imbalance effects. Kappa quantifies the level of agreement between model predictions and ground truth. All experiments were repeated five times with distinct random seeds.

3) Comparison Methods: To verify the effectiveness of the method proposed in this article, six comparative methods were selected for comparison, including ExViT [17], FrIT [50], S^{2}cMamba [53], MFT [51], CCRNet [11], and NCGLF^2 [52]. All comparison methods were evaluated using their respective optimal hyperparameters identified through systematic parameter optimization, ensuring equitable comparison.

To systematically evaluate the proposed method, we compared against three dominant deep learning paradigms (i.e., CNN-based, Transformer-based, and Mamba-based). Specifically, for CNN-based approaches, we selected CCRNet, which employs cascaded convolutional layers to extract hierarchical features and leverages a cross-reconstruction module to enable intermodal information exchange, thereby learning compact fused feature representations through channelwise correlation constraints.

ExViT and MFT were selected for comparative analysis as Transformer-based methods. MFT introduces a multihead cross-patch attention that integrates complementary information from auxiliary modalities as external classification tokens, thereby enhancing encoder representations. ExViT employs parallel branches of position-shared ViTs with integrated separable convolution blocks, where convolutions extract spatial and channel information.

For Mamba-based methods, S^{2}cMamba was selected as the representative approach. S^{2}cMamba introduces a dual branch comprising spectral Mamba and spatial Mamba for feature extraction. The dual branch incorporates the Cross-Mamba module to dynamically fuse multimodal features through bidirectional state-space interactions, enhancing feature richness.

In addition, NCGLF^2 employs CNNs to extract high-frequency features and utilizes Transformer architectures to capture low-frequency information, achieving synergistic integration of localized details and global contextual patterns. These baselines cover the current spectrum of state-space, attention, and convolution approaches in remote sensing classification.

To validate the efficacy of this fusion framework, we benchmarked against a channel-concatenation baseline method termed concat dendritic network (CDN). This approach integrates the two modalities solely through channelwise concatenation before feature extraction, while maintaining identical feature extraction architectures.

C. Comparison Result and Analysis

In this section, we conducted comprehensive experiments on four benchmark datasets to validate the effectiveness of the proposed method through comparisons with state-of-the-art (SOTA) deep learning-based approaches.

Tables V–VIII present the classification performance across four benchmark datasets, with optimal results for each category highlighted in bold to facilitate comparative analysis of method effectiveness. Sections III-C1–III-C4 provide detailed analyses on each dataset.

1) Houston Dataset: Experimental results demonstrate that UDM significantly outperforms all comparative methods across all evaluation metrics, achieving consistent improvements of 4% in OA, 5% in AA, and 4% in Kappa coefficient, as shown in Table V. Specifically, UDM achieves an OA of 91.13% $\pm$ 0.98%, AA of 90.90% $\pm$ 0.50%, and Kappa of 90.42% $\pm$ 1.05%, outperforming CDN by margins of 2.3% in OA, 2.5% in AA, and 2.3% in Kappa. The results validate its enhanced capacity for cross-modal feature integration.

UDM's superiority is also reflected in its per-class accuracy, where it achieves the highest classification rates in 10 of 15 classes. Compared to baselines such as NCGLF^2 and S^{2}cMamba, UDM consistently achieves better results with

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TABLE V
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE HOUSTON DATASET (OA, AA, KAPPA, AND CLASSWISE ACCURACY). BEST RESULTS IN BOLD

Class	ExViT [17]	FrIT [50]	MFT [51]	CCRNet [11]	NCGLF^2 [52]	S^{2}cMamba [53]	**CDN**	**UDM**
1	85.80 ± 3.90	89.69 ± 5.79	91.70 ± 4.63	86.27 ± 0.54	89.24 ± 6.93	82.98 ± 4.36	91.68 ± 4.56	93.38 ± 4.35
2	94.55 ± 1.98	96.62 ± 2.21	89.18 ± 4.70	94.88 ± 0.68	96.34 ± 2.62	77.97 ± 8.97	89.45 ± 3.76	94.03 ± 4.62
3	96.18 ± 2.59	99.70 ± 0.32	98.99 ± 0.61	93.66 ± 2.44	98.66 ± 1.06	97.43 ± 1.95	97.12 ± 2.06	98.32 ± 0.50
4	88.63 ± 2.62	92.47 ± 3.39	95.52 ± 1.95	91.79 ± 0.84	93.27 ± 3.82	91.20 ± 2.77	92.10 ± 2.84	91.89 ± 5.85
5	99.44 ± 0.45	93.26 ± 8.60	97.35 ± 2.11	95.04 ± 0.48	100.00 ± 0.00	97.02 ± 1.14	98.55 ± 1.22	99.59 ± 0.55
6	81.66 ± 7.59	60.40 ± 9.08	60.68 ± 6.45	0.99 ± 0.50	88.32 ± 5.82	68.19 ± 13.14	74.99 ± 9.11	78.80 ± 4.08
7	90.18 ± 5.56	88.87 ± 2.41	88.79 ± 2.13	83.72 ± 1.10	79.50 ± 5.73	78.58 ± 13.60	86.00 ± 4.46	90.04 ± 6.48
8	89.89 ± 6.80	87.68 ± 2.23	91.94 ± 2.70	81.97 ± 0.03	79.72 ± 5.40	84.13 ± 7.75	87.44 ± 5.81	83.26 ± 4.99
9	83.59 ± 5.61	83.55 ± 5.11	72.39 ± 1.55	61.27 ± 2.63	66.09 ± 10.13	76.48 ± 10.01	83.97 ± 5.86	85.56 ± 4.58
10	77.01 ± 15.62	66.35 ± 8.81	87.93 ± 3.53	90.76 ± 4.35	87.82 ± 11.45	78.60 ± 4.46	86.94 ± 8.19	90.75 ± 9.50
11	89.82 ± 3.02	89.95 ± 1.75	85.76 ± 1.59	82.62 ± 0.58	87.3 ± 6.70	94.83 ± 2.93	82.36 ± 8.58	85.77 ± 8.42
12	88.06 ± 11.46	77.42 ± 5.74	86.77 ± 4.08	73.82 ± 0.79	78.04 ± 10.81	78.27 ± 3.58	89.86 ± 7.72	92.92 ± 6.19
13	68.41 ± 9.02	73.31 ± 9.78	65.60 ± 10.72	35.79 ± 5.05	77.61 ± 7.08	68.23 ± 4.40	67.71 ± 34.02	80.05 ± 13.03
14	99.71 ± 0.47	99.19 ± 0.11	96.44 ± 2.52	74.93 ± 10.81	100 ± 0	91.88 ± 4.20	98.38 ± 2.48	99.71 ± 0.36
15	95.97 ± 5.82	98.89 ± 1.46	99.59 ± 0.58	95.35 ± 3.43	99.72 ± 0.55	98.46 ± 1.23	99.14 ± 1.00	99.52 ± 0.48
OA(%)	88.88 ± 0.71	87.14 ± 0.82	88.62 ± 0.54	81.54 ± 0.53	87.16 ± 1.27	84.61 ± 1.82	89.02 ± 0.76	**91.13 ± 0.98**
AA(%)	88.59 ± 0.59	86.49 ± 0.11	87.24 ± 1.07	76.19 ± 0.76	88.11 ± 1.28	84.28 ± 1.93	88.38 ± 1.40	**90.90 ± 0.50**
Kappa(×100)	87.98 ± 0.77	86.08 ± 0.88	87.69 ± 0.59	80.00 ± 0.58	86.11 ± 1.38	83.36 ± 1.97	88.13 ± 0.82	**90.42 ± 1.05**

TABLE VI
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE BERLIN DATASET (OA, AA, KAPPA, AND CLASSWISE ACCURACY). BEST RESULTS IN BOLD

Class	ExViT [17]	FrIT [50]	MFT [51]	CCRNet [11]	NCGLF^2 [52]	S^{2}cMamba [53]	**CDN**	**UDM**
1	79.63 ± 11.70	79.16 ± 3.39	73.42 ± 1.63	78.40 ± 2.40	84.97 ± 5.51	75.91 ± 5.30	79.19 ± 6.46	83.58 ± 5.40
2	93.61 ± 2.66	92.46 ± 1.22	94.49 ± 0.21	91.93 ± 0.44	92.97 ± 2.79	88.24 ± 8.12	94.85 ± 1.50	93.72 ± 2.38
3	31.78 ± 16.85	38.46 ± 8.43	56.56 ± 3.58	14.56 ± 0.88	34.21 ± 23.38	41.05 ± 15.94	49.36 ± 26.25	55.26 ± 25.44
4	75.32 ± 15.07	81.42 ± 3.41	82.22 ± 3.31	75.73 ± 1.25	82.55 ± 3.48	83.51 ± 9.19	86.93 ± 2.33	85.43 ± 3.98
5	69.79 ± 16.70	85.35 ± 6.04	80.83 ± 4.89	80.66 ± 7.16	79.31 ± 4.88	80.32 ± 8.42	72.86 ± 11.21	73.19 ± 9.10
6	5.49 ± 10.46	29.01 ± 10.29	29.22 ± 16.45	7.86 ± 2.86	21.03 ± 12.76	24.77 ± 26.16	39.50 ± 8.44	42.83 ± 12.12
7	36.41 ± 7.96	27.22 ± 4.19	14.03 ± 1.62	19.05 ± 2.07	30.91 ± 16.94	29.61 ± 18.58	23.82 ± 13.27	26.01 ± 9.36
8	37.43 ± 25.18	64.68 ± 16.96	63.67 ± 8.08	37.76 ± 10.24	23.99 ± 19.65	22.87 ± 23.91	16.45 ± 14.96	32.93 ± 21.74
OA(%)	79.82 ± 3.05	81.30 ± 0.74	81.78 ± 0.16	77.61 ± 0.97	81.42 ± 1.36	78.10 ± 3.71	82.86 ± 1.00	**83.24 ± 0.77**
AA(%)	53.68 ± 8.14	**62.22 ± 3.00**	61.81 ± 2.98	50.74 ± 2.88	56.24 ± 2.47	55.79 ± 4.67	57.87 ± 2.50	61.62 ± 2.32
Kappa(×100)	65.56 ± 7.32	69.18 ± 1.18	69.39 ± 0.35	62.03 ± 1.81	69.54 ± 1.83	65.06 ± 3.93	71.65 ± 1.69	**72.73 ± 0.88**

smaller performance variance, indicating its robustness to random initialization and training fluctuations.

2) Berlin Dataset: As evidenced by the quantitative results in Table VI for the Berlin dataset, the UDM framework achieves SOTA performance, surpassing all comparative methods across key evaluation metrics (OA/Kappa).

Compared to the previous best method in AA (FrIT with 62.22%), UDM offers comparable performance while achieving considerably higher OA and Kappa, indicating a better tradeoff between per-class balance and overall classification quality. In addition, the standard deviations of UDM across all three metrics are relatively low, indicating stability and robustness across repeated runs. Compared to earlier models such as MFT (OA = 81.78%) and CCRNet (OA = 77.61%), UDM achieves clear and consistent improvements. Its combination of high OA and balanced AA also confirms that it not only classifies dominant categories well, but also handles minority or more ambiguous classes effectively.

3) MUUFL Dataset: Through extensive experiments on the MUUFL dataset, this model demonstrates superior classification performance, achieving an OA of 86.74% ± 1.26%, which surpasses all competing methods by a significant margin of at least 2.8%. The second-best performer, CDN, achieves 84.91% ± 0.92%, further validating the effectiveness of the DNL and fusion mechanism.

In terms of AA, which evaluates performance across all classes regardless of sample size, CDN achieves the best

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TABLE VII

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE MUUFL DATASET (OA, AA, KAPPA, AND CLASSWISE ACCURACY). BEST RESULTS IN BOLD

Class	ExViT [17]	FrIT [50]	MFT [51]	CCRNet [11]	NCGLF^2 [52]	S^{2}cMamba [53]	**CDN**	**UDM**
1	95.70 ± 1.36	96.18 ± 1.92	95.91 ± 1.66	95.35 ± 0.05	93.21 ± 2.42	94.61 ± 1.44	95.65 ± 1.21	96.98 ± 0.75
2	66.47 ± 8.37	73.34 ± 7.92	66.38 ± 12.95	64.78 ± 4.15	73.25 ± 13.32	79.24 ± 7.44	78.55 ± 4.96	80.91 ± 6.64
3	71.73 ± 12.16	74.66 ± 2.47	76.40 ± 9.62	75.90 ± 0.56	70.56 ± 9.83	69.46 ± 10.15	77.89 ± 5.01	80.05 ± 4.68
4	41.70 ± 8.56	75.43 ± 9.13	43.01 ± 18.54	57.33 ± 12.97	55.23 ± 27.51	58.14 ± 21.77	43.68 ± 19.81	48.23 ± 31.09
5	89.77 ± 2.37	84.62 ± 6.76	87.25 ± 2.90	83.82 ± 2.52	81.23 ± 8.51	90.78 ± 2.76	92.32 ± 1.22	92.94 ± 1.09
6	6.78 ± 9.21	7.12 ± 14.23	11.30 ± 14.63	55.91 ± 0.26	23.31 ± 38.29	17.58 ± 32.76	3.31 ± 5.06	5.12 ± 9.52
7	59.11 ± 11.17	72.89 ± 9.16	61.10 ± 10.54	31.94 ± 20.15	65.34 ± 8.62	65.75 ± 5.05	65.85 ± 11.08	69.03 ± 10.01
8	94.31 ± 1.44	81.53 ± 9.54	91.47 ± 3.93	86.88 ± 0.40	90.91 ± 3.51	93.23 ± 3.41	91.58 ± 5.08	91.30 ± 3.36
9	34.27 ± 8.11	25.81 ± 7.61	30.68 ± 19.07	7.72 ± 0.84	12.50 ± 8.16	19.04 ± 10.36	23.12 ± 7.12	17.67 ± 9.48
10	2.15 ± 2.82	10.91 ± 8.74	2.28 ± 3.68	4.92 ± 4.31	1.44 ± 1.80	5.81 ± 6.86	2.50 ± 3.66	0.63 ± 0.97
11	14.75 ± 20.36	26.75 ± 25.18	11.48 ± 19.22	8.57 ± 29.28	33.79 ± 32.65	46.43 ± 26.08	22.74 ± 23.05	26.02 ± 21.26
OA(%)	83.28 ± 0.69	83.66 ± 2.18	83.35 ± 1.36	81.00 ± 0.33	81.40 ± 1.50	84.24 ± 0.85	85.53 ± 1.15	**86.74 ± 1.22**
AA(%)	52.43 ± 4.13	57.20 ± 3.50	52.48 ± 5.47	56.20 ± 2.19	54.62 ± 6.54	**58.19 ± 4.24**	54.29 ± 1.78	55.35 ± 3.48
Kappa(×100)	77.49 ± 1.06	78.04 ± 2.88	77.57 ± 1.89	74.41 ± 0.48	75.31 ± 1.87	78.87 ± 1.19	80.46 ± 1.48	**81.94 ± 1.79**

TABLE VIII

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE WUHAN DATASET (OA, AA, KAPPA, AND CLASSWISE ACCURACY). BEST RESULTS IN BOLD

Class	ExViT [17]	FrIT [50]	MFT [51]	CCRNet [11]	NCGLF^2 [52]	S^{2}cMamba [53]	**CDN**	**UDM**
1	86.84 ± 3.20	85.86 ± 2.37	84.69 ± 2.93	79.30 ± 2.44	81.27 ± 5.09	88.78 ± 2.23	92.31 ± 1.16	93.79 ± 0.44
2	28.94 ± 11.90	34.87 ± 14.36	41.87 ± 10.35	10.19 ± 5.62	43.82 ± 9.91	32.61 ± 11.34	42.02 ± 7.09	48.75 ± 12.14
3	46.93 ± 7.65	42.75 ± 5.27	44.11 ± 11.59	32.43 ± 7.57	46.35 ± 18.95	47.92 ± 9.92	62.70 ± 8.67	62.16 ± 7.81
4	14.73 ± 14.24	29.45 ± 9.42	25.43 ± 10.42	12.27 ± 4.82	26.05 ± 19.03	30.19 ± 9.70	43.99 ± 11.53	45.57 ± 13.17
5	0.38 ± 0.77	4.70 ± 4.10	7.57 ± 7.65	0.03 ± 0.04	4.19 ± 8.13	7.52 ± 9.41	17.95 ± 9.20	22.03 ± 24.36
6	3.82 ± 5.25	9.43 ± 8.27	9.28 ± 3.93	0.82 ± 0.64	1.31 ± 1.11	9.96 ± 4.76	22.47 ± 9.24	26.78 ± 12.26
7	80.44 ± 4.12	71.32 ± 4.80	70.89 ± 5.18	76.73 ± 6.65	81.35 ± 6.76	77.94 ± 2.11	88.46 ± 5.12	86.11 ± 5.04
8	0.96 ± 1.36	7.58 ± 2.46	9.89 ± 7.94	0.00 ± 0.00	1.12 ± 2.13	5.98 ± 2.40	10.56 ± 5.11	5.54 ± 10.17
9	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
10	79.17 ± 2.36	78.17 ± 3.67	77.67 ± 8.15	65.67 ± 3.19	79.56 ± 5.73	75.18 ± 3.37	86.97 ± 3.05	88.74 ± 2.61
11	68.01 ± 5.49	55.24 ± 10.16	55.00 ± 12.48	51.93 ± 3.23	62.83 ± 7.44	61.13 ± 10.59	71.79 ± 8.10	68.10 ± 11.97
12	10.09 ± 10.31	16.80 ± 6.31	27.01 ± 16.13	3.35 ± 2.59	4.75 ± 5.11	14.67 ± 5.45	37.60 ± 10.25	37.97 ± 9.27
13	0.24 ± 0.36	4.06 ± 3.22	4.04 ± 3.21	0.01 ± 0.02	2.07 ± 4.10	2.94 ± 3.54	6.07 ± 4.29	8.85 ± 9.89
OA(%)	60.71 ± 1.26	58.39 ± 3.07	58.90 ± 1.54	50.84 ± 0.58	60.87 ± 1.90	60.89 ± 1.80	71.05 ± 1.50	**71.33 ± 2.13**
AA(%)	32.35 ± 0.90	33.86 ± 2.23	35.19 ± 1.08	25.59 ± 0.35	33.44 ± 1.84	34.99 ± 1.76	44.84 ± 0.67	**45.72 ± 2.28**
Kappa(×100)	53.64 ± 1.38	51.38 ± 3.52	52.28 ± 1.49	41.92 ± 0.47	54.11 ± 2.19	54.28 ± 1.94	66.12 ± 1.63	**66.52 ± 2.31**

result with 58.43% ± 1.99%, outperforming all other methods. This further substantiates the exceptional feature extraction capacity of DNL architectures. Additionally, the Kappa coefficient of 81.94% ± 1.79% further confirms the robustness of UDM's predictions and its strong agreement with ground truth labels.

4) Wuhan Dataset: As presented in Table VIII, the proposed UDM framework demonstrates exceptional performance on the challenging Wuhan dataset, securing state-of-the-art results across all primary evaluation metrics. UDM achieves an OA of 71.33% ± 2.13%, an AA of 45.72% ± 2.28%, and a Kappa coefficient of 66.52% ± 2.31%, surpassing all comparative baselines. Notably, both UDM and its variant CDN significantly outperform the nearest competitor, S^{2}cMamba, by a remarkable margin of over 10% in OA, highlighting the model's superior capability in handling large-scale, cross-region environments.

In terms of per-class performance, UDM achieves the highest accuracy in the majority of categories. However, it is observed that Class 9 yields a classification accuracy of zero across all methods. This is attributed to the experimental setup where only 200 samples were randomly selected for training. Due to the scarcity or uneven distribution of Class 9, this class was likely underrepresented or absent in the training subset, making it unlearnable under the current few-shot constraints. Despite this limitation, UDM maintains the highest AA, further validating the effectiveness of the DNL and uncertainty-based mechanism in mitigating the impact of class imbalance.

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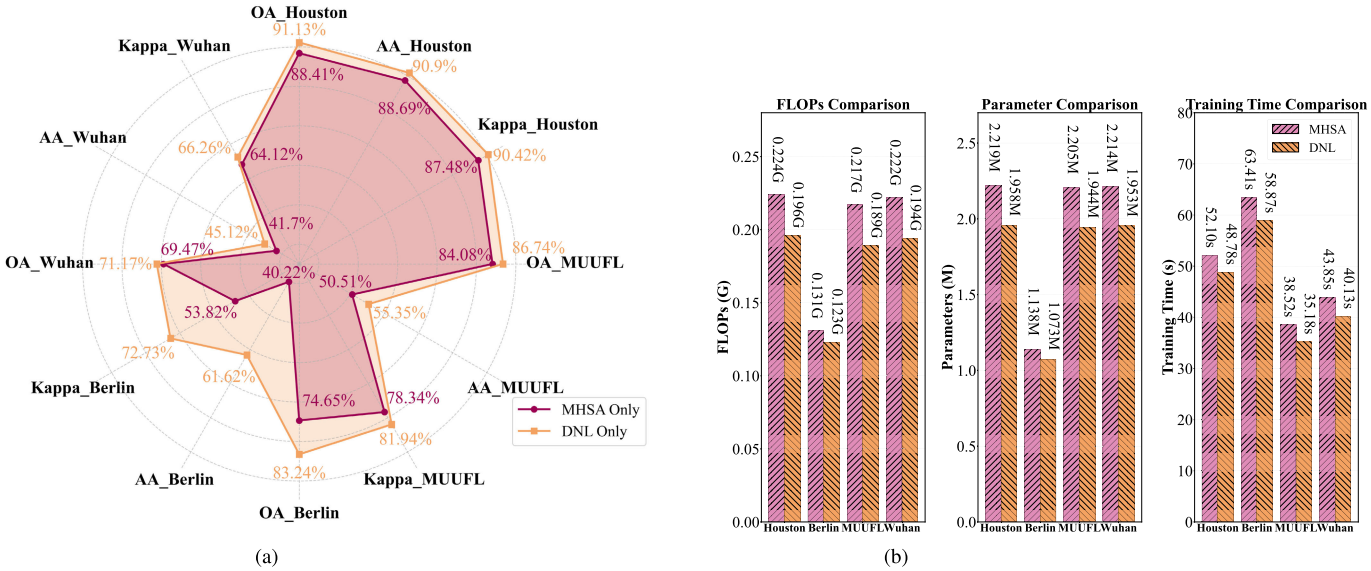


Fig. 7. Quantitative comparison between DNL and MHSA on the four dataset. DNL achieves higher accuracy with lower computational costs. (a) Performance comparison: OA, AA, and Kappa. (b) Efficiency comparison: FLOPs, parameters, and training time.

TABLE IX

QUANTITATIVE COMPARISON OF CLASSIFICATION PERFORMANCE BETWEEN THE PROPOSED FRAMEWORKS (ULM AND UDM)

Method	Houston			Berlin			MUUFL		
	OA(%)	AA(%)	Kappa(×100)	OA(%)	AA(%)	Kappa(×100)	OA(%)	AA(%)	Kappa(×100)
ULM	90.80	90.53	90.06	81.08	57.71	68.79	86.71	**56.35**	81.87
UDM	**91.13**	**90.90**	**90.42**	**83.24**	**61.62**	**72.73**	**86.74**	55.35	**81.94**

D. Ablation Study

To verify the effectiveness of DNL in feature transformation, as well as the improvements in classification accuracy and computational efficiency when it is used for feature extraction, this article conducted two sets of ablation experiments. First, to validate the generality of DNL in feature extraction, comparisons were conducted with multihead self-attention (MHSA) mechanisms. Second, DNL was compared with linear layers to verify the effectiveness of feature transformation.

1) Dendritic Feature Capacity: To verify the generality of the DNL proposed in this article in the feature extraction stage, we replaced the DNL used for feature extraction in the model with MHSA mechanisms to verify the performance of DNL. Fig. 7 delineates the classification performance across four benchmark datasets utilizing different feature abstraction architectures. The quantitative comparisons reveal that dendritic layers extract more discriminative feature representations than MHSA, consequently achieving superior scores across all three evaluation metrics (OA, AA, and Kappa). Notably, the dendritic architecture demonstrated 11.8% fewer parameters and 9.77% reduced computational overhead (measured in FLOPs) compared to the MHSA baseline, alongside an average 7.67% reduction in training time, while maintaining enhanced feature quality.

2) Compared With Linear: Motivated by the superior nonlinear computational capacities inherent in dendritic neuron, in order to validate the effectiveness of the DNL, we conducted ablation studies by replacing all DNL in UDM with standard linear layers, resulting in a modified architecture termed ULM. As reported in Table IX, under the same training parameter configurations, the UDM incorporating DNL demonstrates superior performance over the ULM with conventional linear layers across both the Houston and Berlin datasets, while achieving comparable experimental results on the MUUFL dataset. This demonstrates that DNL effectively enhances feature transformation and extraction capabilities.

E. Training Efficiency

To further verify the efficiency of the method in this article, we compared the training time of each method (the training time was obtained when each algorithm achieved its best accuracy). The time consumption of ExViT, FrIT, S^2cMamba, and the proposed UDM is shown in Fig. 8. These methods include Transformer-based and Mamba-based approaches. From Fig. 8, it can be seen that the proposed method has faster training speed in MRSC, even though Mamba-based methods are known for their speed. This approach achieves a balance between classification accuracy and training speed.

F. Parameter Discussion

1) Different Dimensions of d_f: The size of the feature dimension d_f plays a crucial role in determining the total number of model parameters and significantly affects the

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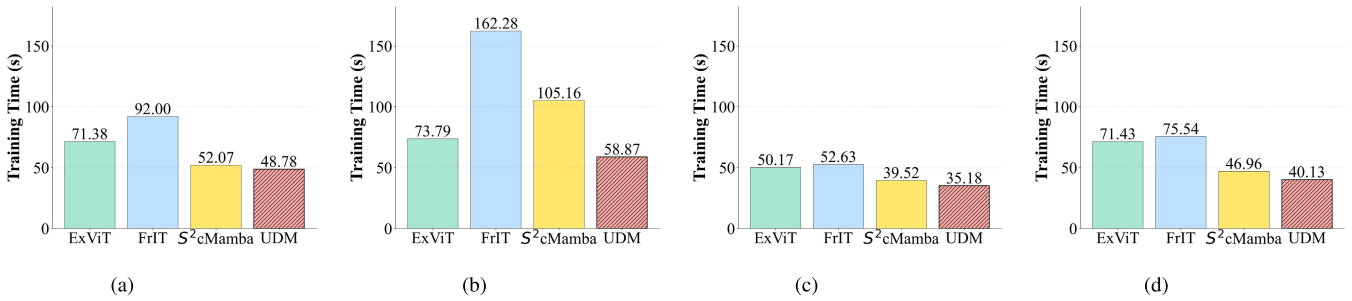


Fig. 8. Training time comparison across methods (i.e., ExViT, FrIT, S^{2}cMamba, and the proposed UDM) on the four benchmark datasets. (a) Houston. (b) Berlin. (c) MUUFL. (d) Wuhan.

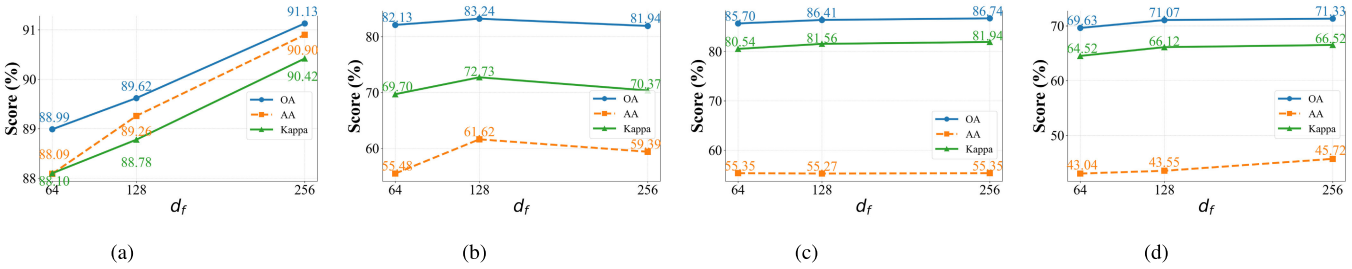


Fig. 9. Quantitative results for different dimension d_f. (a) Houston. (b) Berlin. (c) MUUFL. (d) Wuhan.

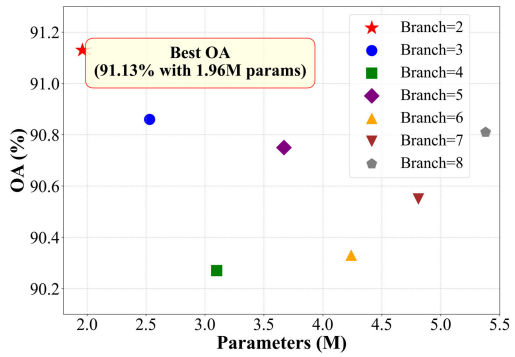


Fig. 10. Impact of the number of dendritic branches on parameters and accuracy.

TABLE X
ABLATION STUDY OF LOSS FUNCTIONS ON THE MUUFL DATASET

L_{FE}	✗	✓	✓	✗	✓
$L_{KLgauss}$	✓	✗	✓	✗	✓
$L_{KLcross}$	✓	✓	✗	✗	✓
OA(%)	86.59	82.25	86.55	82.16	**86.74**
AA(%)	55.01	**56.69**	54.99	54.67	55.35
Kappa($\times$100)	81.70	76.50	81.67	76.19	**81.94**

computational speed. Consequently, we conducted multiple experiments on four different datasets to explore the impact of d_f setting and observed how it affects performance on each dataset. As shown in Fig. 9, when d_f is set to 256, the model performs best on the Houston, MUUFL and Wuhan datasets, indicating that a larger d_f can help capture more complex features. For the Berlin dataset, however, the best performance is achieved when d_f is set to 128, suggesting that this setting offers a good tradeoff between accuracy and efficiency, likely due to the unique characteristics of the dataset.

2) Function of Loss Functions: This section aims to validate the efficacy of the proposed loss functions in the model including L_{KLgauss}, L_{KLcross}, and L_{FE}. To ensure fair comparison, the cross-entropy (CE) loss remains unchanged across all ablation studies, as CE is the fundamental loss function for classification tasks whose necessity is well-established.

Table X presents the performance of different loss functions on the MUUFL dataset. Ablation studies demonstrate that L_{KLgauss} (the Gaussian prior loss constraining both unimodal enhanced feature and fusion feature) serves as the core regularization constraint. The removal of L_{KLgauss} results in a notable performance degradation, with the OA decreasing by 4.49% from 86.74% to 82.25%, and the Kappa coefficient dropping by 5.44% from 81.94% to 76.50%. This result confirms the effectiveness of L_{KLgauss} in enhancing model performance through regularization. In contrast, the contrastive loss L_{FE} for unimodal enhancement and cross-modal distribution loss L_{KLcross} play auxiliary roles. When L_{FE} and L_{KLcross} are individually removed, OA reduction is less than 0.2% (86.59% versus 86.74% and 86.55% versus 86.74%, respectively). Given their originally intended purposes, the results suggest that the application of Gaussian priors enhances the overall performance of the model.

3) Different Dendritic Branch Number of M: The dendritic branch number M has a significant impact on the performance of the DNL as well as its parameter complexity. To investigate

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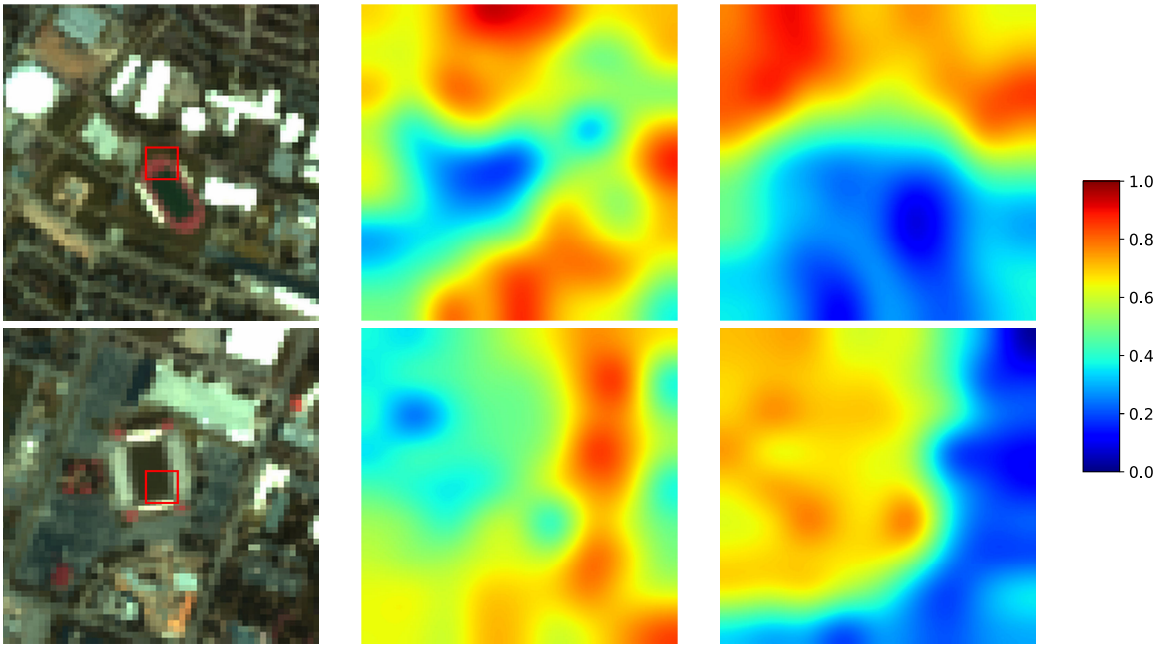


Fig. 11. Visualization of structural interpretability. The distinct activation patterns of different dendritic branches demonstrate that the DNL learns complementary features in different subspaces. (a) Input patch. (b) Branch 1 activation. (c) Branch 2 activation.

the impact of dendritic branch quantity on the modeling capability of the dendritic layer, we conducted comparative experiments with varying numbers of branches. As shown in Fig. 10, increasing the number of branches leads to a significant rise in model parameters, while the OA remains relatively stable. This observation suggests that the current task does not heavily rely on increased model capacity. However, there is potential to further exploit the model's capability in more complex tasks or larger scale datasets. More importantly, the results highlight the proposed dendritic layer's ability to maintain stable performance under increased architectural complexity, demonstrating excellent scalability and robustness. These findings lay a solid foundation for future applications in multitask learning, high-dimensional modality fusion, or fine-grained classification, and reveal broad prospects and research potential for further exploration.

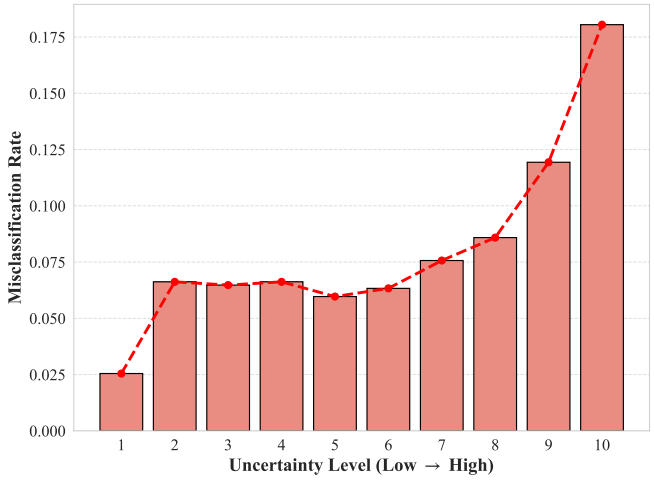


Fig. 12. Correlation between aleatoric uncertainty and error rate. The increasing trend verifies that high uncertainty effectively indicates a higher probability of misclassification, providing a quantifiable metric for interpretability.

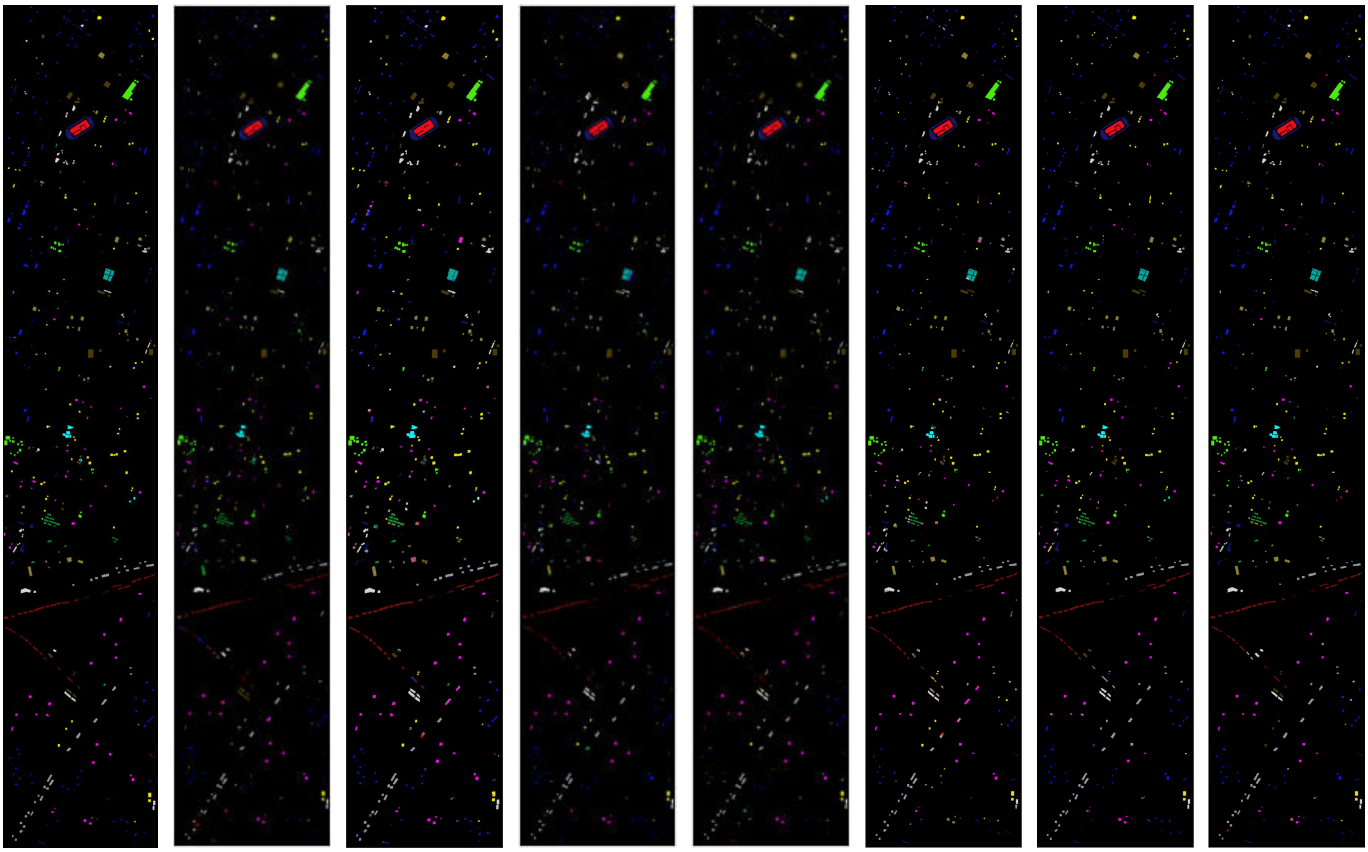
G. Model Interpretability

To demonstrate the transparency of our proposed UDM, we conducted an analysis from two perspectives: structural feature and uncertainty-based reliability.

1) Visualizing Dendritic Branch Analysis: In Section II-B, we claimed that the DNL performs subspace-local computations. To verify this mechanism, we conducted a visualization experiment using the Houston dataset as a representative case. We specifically extracted the intermediate activation maps from the synaptic–dendritic computation stage of the last DNL layer within the feature extraction module. Since each neuron in this layer consists of multiple branches, we visualized the outputs of two distinct branches prior to their aggregation at the membrane layer. As illustrated in Fig. 11, for a given input sample, the two dendritic branches exhibit distinct activation patterns. Branch 1 tends to capture high-frequency local details with scattered activation peaks, whereas Branch 2 focuses on broader regional semantic consistency, showing a clear spatial separation between distinct spatial subregions. This contrast confirms that the proposed DNL successfully captures distinct feature subrepresentations of the same input, demonstrating the strong interpretability of the structure.

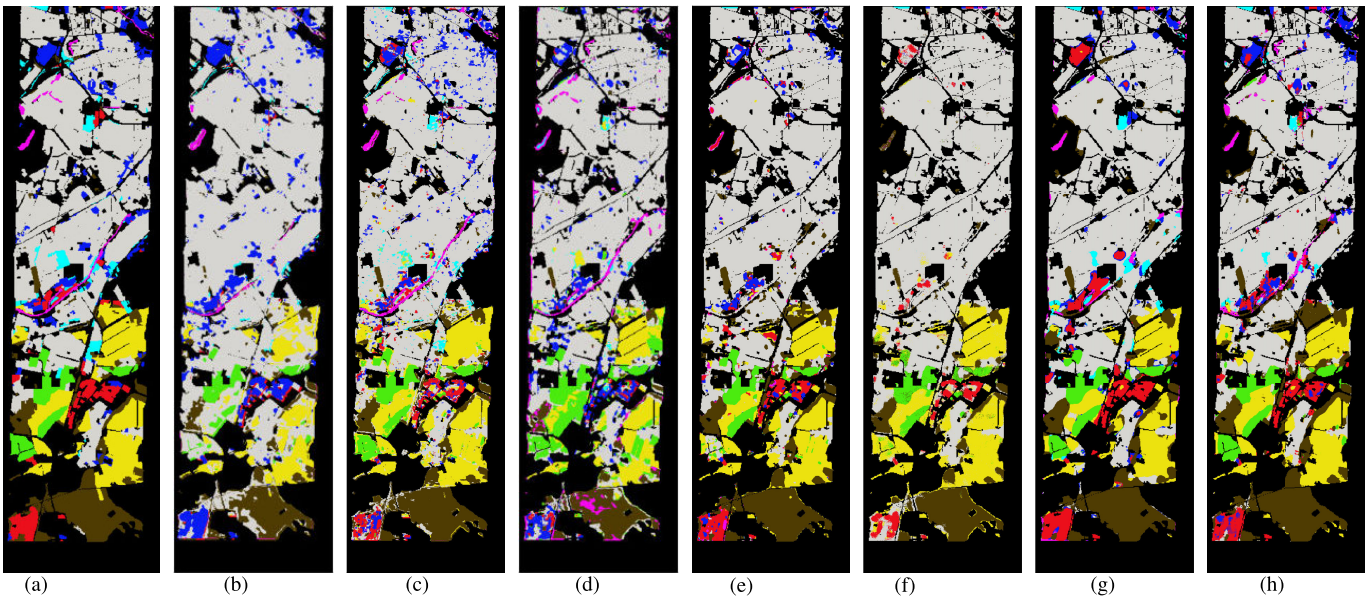
2) Interpretability via Uncertainty Quantification: The quantified aleatoric uncertainty σ^m serves as key interpretability metrics for the decision-making process. Thereby, we examine the reliability of the uncertainty estimation. We define the uncertainty score as the total predicted variance $\sum \sigma^m$

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(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 13. Classification maps generated by different methods on the Houston dataset. (a) Ground truth. (b) ExViT (88.88%). (c) FrIT (87.14%). (d) MFT (88.62%). (e) NCGLF^2 (87.16%). (f) S^{2}cMamba (84.61%). (g) CDN (89.02%). (h) UDM (91.13%).



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 14. Classification maps generated by different methods on the Berlin dataset. (a) Ground truth. (b) ExViT (79.82%). (c) FrIT (81.30%). (d) MFT (81.78%). (e) NCGLF^2 (81.42%). (f) S^{2}cMamba (78.10%). (g) CDN (82.86%). (h) UDM (83.24%).

derived from the uncertainty-aware fusion module, serving as a metric for model interpretability. Using the test set of the Houston dataset, we collected the uncertainty scores and prediction results for all samples. Fig. 12 illustrates the strong positive correlation between the estimated aleatoric uncertainty and the classification error rate. To quantify this correlation, the test samples were sorted by their uncertainty scores and divided into ten equal-frequency intervals. As

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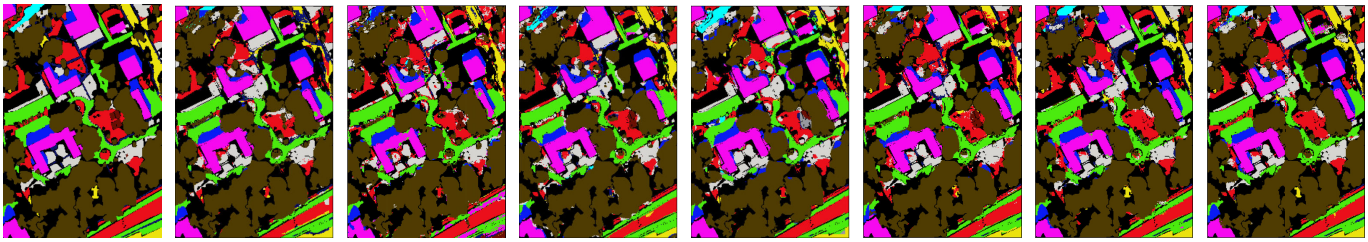


Fig. 15. Classification maps generated by different methods on the MUUFL dataset. (a) Ground truth. (b) ExViT (83.28%). (c) FrIT (83.66%). (d) MFT (83.35%). (e) NCGLF^2 (81.40%). (f) S^{2}cMamba (84.24%). (g) CDN (85.53%). (h) UDM (86.74%).

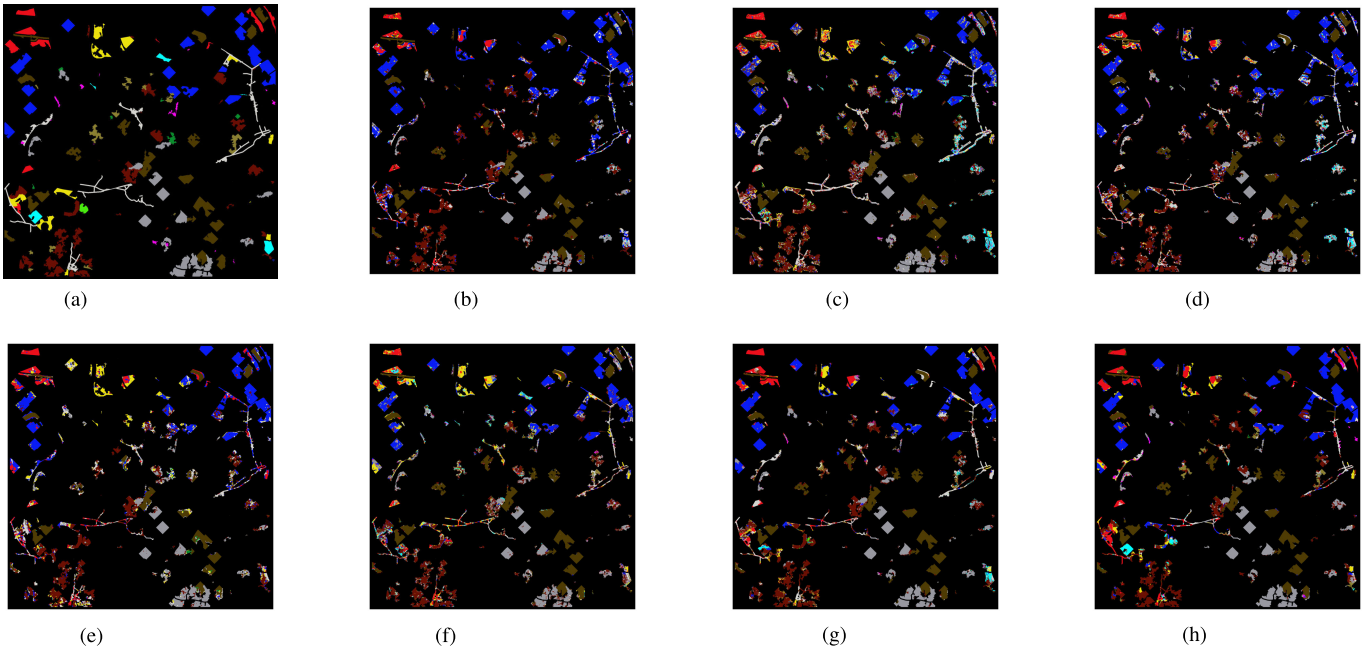


Fig. 16. Classification maps generated by different methods on the Wuhan dataset. (a) Ground truth. (b) ExViT (60.71%). (c) FrIT (58.39%). (d) MFT (58.90%). (e) NCGLF^2 (60.87%). (f) S^{2}cMamba (60.89%). (g) CDN (71.05%). (h) UDM (71.33%).

shown in Fig. 12, the samples in the lowest uncertainty achieve nearly perfect classification, while the error rate rises sharply in the highest uncertainty. This demonstrates that our uncertainty-aware module effectively captures the confidence of predictions, thereby confirming that the aleatoric uncertainty $\boldsymbol{\sigma}^m$ serves as an effective and interpretable metric for assessing decision trustworthiness.

H. Visual Comparison

Figs. 13–16 present the classification visualization results. Specifically, Figs. 13(a) and 16(a) provide the ground truth reference. Figs. 13(b)–(h)–16(b)–(h) illustrate the classification results obtained by different methods including ExViT, FrIT, MFT, NCGLF^2, S^{2}cMamba, the proposed CDN, and UDM. As shown in Figs. 13–16(g) and (h), the proposed methods achieve superior classification performance compared with other methods, and produce clearer object boundaries. Specifically, Fig. 14(b)–(g) reveals that in the Berlin dataset, the regions located in the lower-left and upper-left corners are prone to confusion among residential area, industrial area, and commercial area. In the ground truth, the lower-left region marked in red corresponds to industrial area, while the upper-left region marked in blue represents commercial area. Compared with other competing methods, the proposed method, as illustrated in Fig. 14(h), introduces fewer noisy predictions. Specifically, most of the lower-left region is correctly classified as industrial area which is colored red, and the upper-left region is correctly classified as commercial area which is colored blue. Similar conclusions can also be drawn on the Houston and MUUFL datasets. The effectiveness of the proposed method stems from its ability to fuse complementary features from different modalities, which reduces pixel noise in the classification maps and produces results that are more consistent with the ground truth.

IV. Conclusion

In this article, we explore an UDM for multimodal remote sensing image classification. To achieve interpretability and nonlinear capability, UDM introduces the DNL, which is designed by mimicking biological mechanisms. Specifically, DNL projects inputs onto dendritic branches via independent sets of synaptic weights, modulates these projected inputs

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using branch-specific biases, and integrates the resulting signals through membrane-level weighted aggregation. During the single-modality feature extraction stage, a DSCE module incorporating DNL is first designed to extract stable unimodal features through local computation processes, while reducing parameters by 11.8% and FLOPs by 9.77% compared to the Transformer. Building upon the single-modal features obtained from DSCE, the DUFE module strengthens the representations of each modality by explicitly quantifying their aleatoric uncertainty. At the multimodal feature fusion stage, DUAF module adopts a dynamic fusion strategy to balance intermodal consistency and redundancy. As a whole, the superiority of UDM lies in its uncertainty-aware ability and competitive performance in extensive experiments on Houston, Berlin, MUUFL, and Wuhan datasets show that it outperforms SOTA methods based on Transformer, CNN, and Mamba, with ablation studies validating the necessity of DSCE, DUFE, and DUAF.

In the future, efforts will concentrate on deeply combining interpretable dendritic structures with various feature fusion strategies, as well as optimizing the internal computation logic of these structures. We will also aim to further reduce the model's parameter count and computational cost and integrate epistemic uncertainty quantification to improve its adaptability to complex scenes.

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